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The recombination mechanism and defect energy levels of Ag-doped $ZnSnN_2$/$ZnSnN_2$ pn homojunction solar cells

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AFFILIATIONS

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ABSTRACT

The photovoltaic performance of $ZnSnN_2$ homojunction has never been studied. Here, $ZnSnN_2$ homojunction solar cells with Ag-doped $ZnSnN_2$ as the *p*-type layer are fabricated with magnetron sputtering deposition, and the mechanism limiting the photovoltaic performance is revealed. The junction is abrupt. The current transport is dominated by thermionic emission below 320 K and generation-recombination (GR) in the space charge region in 330–380 K. Below 320 K, the short circuit current density and the open circuit voltage are limited by non-radiative interface recombination. The open circuit voltage loss also results from interfacial barrier height inhomogeneity, which obeys the Gaussian distribution model or the temperature fluctuation of interface barrier heights for holes. At higher temperatures, deep energy levels in the middle of the bandgap result in the GR in the space charge region. One shallow energy level at 0.13 eV and one interface energy level at 0.72 eV above the valence band maximum are observed. The former is likely induced by substitutional Ag-doping.

Thin film solar cells have significantly advanced the development of solar energy technologies. To date, numerous promising candidate materials have emerged, including Cu_2ZnSnS_4, Sb_2Se_3, and $ZnSnN_2$ (ZTN).[1–5] ZTN is a ternary nitride semiconductor and is attracting attention due to its tunable direct bandgap and high absorption coefficient within the solar spectrum, non-toxicity, and Earth-abundance of the constituent elements, demonstrating great potential for photovoltaic, photocatalytic, and light-emitting applications.[1,2,6–11]

In early studies, ZTN is found to exhibit degenerate electron concentrations, mainly due to the low formation energies of donor defects, including Sn_{Zn} (Sn on Zn sites) and unintentional O_N (O on N sites), compared to acceptor defects like Zn_{Sn} (Zn on Sn sites).[6–8] Subsequent methods, including non-stoichiometric growth with higher Zn content and hydrogen doping or passivation, are effective in reducing the electron concentration to the magnitude of $10^{16}\,cm^{-3}$.[9,12–16] Various ZTN-based devices, such as Schottky and heterojunction solar cells and photodetectors, have been fabricated.[17–22] More recently, its *p*-type conduction has been achieved with doping Li, Ba, Al, Ga, or Ag[3,23–25] and rectifying ZTN pn homojunctions[3] with Al- or Ga-doping and no photovoltaic effect have been constructed.

However, ZTN homojunctions with the photovoltaic effect have never been reported. In this work, we demonstrate that ZTN homojunction solar cells with Ag-doped $ZnSnN_2$ (AZTN) as the *p*-type layer can be prepared using magnetron sputtering, which is compatible with industry production. The properties of the solar cells are studied, and the mechanism that limits the open circuit voltage (V_{oc}) and short circuit current density (J_{sc}) is revealed.

To obtain the properties of thin films, K9 glass substrates and Si are used as the substrates. Indium tin oxide (ITO) coated glass is used as the substrate to prepare the homojunctions with the structure of glass\ITO\ZTN\AZTN\Pt. The details about preparing and characterizing the films, including ZTN, AZTN, and Pt and the homojunctions, are in the supplementary material (the steps to prepare the homojunctions are in Fig. S1).

The XRD patterns of the ZTN and AZTN thin films are in Fig. S2, supplementary material. ZTN is wurtzite[7,9] while AZTN is amorphous or nanocrystalline.[26] The surface morphology of the thin films is in Fig. S3. The reflectance (R_e), transmittance (T_r), and Tauc plots are in Fig. S4. The film thickness d for ZTN and AZTN is 334.3 and 66.6 nm, respectively. The optical bandgap E_g for ZTN and AZTN is 2.02 and 1.52 eV,[3,23,24] respectively. The electrical properties of the films are characterized using Hall effect measurements. The ZTN layer is *n*-type conductive with an electron density of 1.51×10^{17} cm^{-3} while the AZTN layer is *p*-type conductive with a hole density of 1.59×10^{15} cm^{-3} and a mobility μ_p of 2.68 cm^2 V^{-1} s^{-1}.

Figure 1(a) shows the dark current density (J) vs the applied voltage (V) curve of the homojunction (glass\ITO\ZTN\AZTN\Pt). The rectification suggests the formation of a p–n junction at the ZTN\AZTN interface since ITO and Pt form Ohmic contacts with the adjacent ZTN and AZTN layers.[25,26] Under illumination, the homojunction exhibits a weak but discernible photovoltaic response [Fig. 1(b)], with a photoelectric conversion efficiency (E_{ff}) of 0.0003%, along with a V_{oc} of 0.21 V, a J_{sc} of 0.0046 mA/cm^2, and a fill factor (FF) of 31.24%.

Figure 1(c) shows the dark forward double-logarithmic J–V curve, which can be divided into three distinct regions:[27] an Ohmic region (slope $\approx$ 1) at low bias voltages, a trap-filled-limited (TFL) region (slope > 2) at intermediate voltages, where the current increases sharply due to the filling of trap states, and a child's law region (slope $\approx$ 2) at higher voltages, where space charge limited current (SCLC) dominates. The transition voltage from the Ohmic to the TFL region is defined as the trap-filled-limit voltage ($V_{TFL} = 0.52$ V). The trap density (N_{trap}) can be estimated[28] since $N_{trap} = 2\varepsilon_r\varepsilon_0 V_{TFL}/(qd^2)$, where $\varepsilon_r = 11^2$ is the relative dielectric constant of ZTN, ε_0 is the vacuum permittivity, q is the elementary charge, and d is the thickness of the AZTN layer. The very possible origin of these traps is deep defects in the bulk. The calculated value of N_{trap} for the homojunction is 1.43×10^{17} cm^{-3}, larger than that (10^{14} cm^{-3})[19] of Cu_2ZnSnS_4\ZTN, resulting in low efficiency.

The capacitance–voltage (C–V) curve measured at 10 kHz is shown in Fig. 1(d). In general, the measured capacitance C of a homojunction consists of the space charge capacitance (C_{SC}) and the interface state capacitance (C_{IS}) in parallel. Therefore, $C = C_{IS} + C_{SC}$.[29] C_{SC} equals[30] $A_{SC}/(V_{bi} - V)^{1/2}$ and $A_{SC} = \{0.5\varepsilon_r\varepsilon_0 N_A N_D/(N_A+N_D)\}^{1/2}$, where N_A and N_D are the doping concentrations in the *p*-type and *n*-type layers, and V_{bi} is the built-in potential. Using C_{IS}, A_{SC}, and V_{bi} as the fitting parameters, the C–V data can be fitted to $C = C_{IS} + A_{SC}/(V_{bi} - V)^{1/2}$ with the least squares methods. The results show that $C_{IS} = 2.0 \times 10^{-7}$ F/cm^2, $A_{SC} = 9.1 \times 10^{-7}$ V$^{0.5}$F/cm^2, and $V_{bi} = 0.66$ V [Fig. 1(e)]. N_A is estimated to be 1.07×10^{17} cm^{-3} since A_{SC} can be simplified to be $\{0.5\varepsilon_r\varepsilon_0 N_A\}^{1/2}$ according to the Hall effect measurement. Therefore, the pn junction is abrupt. In addition, since Hall effect measurements show that $N_D > N_A$, V_{bi} is related to the built-in potential of the holes, as shown in the energy band diagram in Fig. S5.[21,25,31,32]

The temperature-dependent J–V curves of the homojunction are in Figs. S6(a) and S6(b). The current transport mechanism[33,34] is thermionic emission (TE) in 190–320 K and generation recombination (GR) in the space charge region in 330–380 K [Figs. S6(c) and S6(d)]. The Cheungs' method[35] is used to extract the barrier height φ_b, ideality factor n, and series resistance R_s. The barrier height is inhomogeneous and obeys the Gaussian distribution model[36–38] in which the relation between φ_b and temperature T is

$$\varphi_b = \overline{\varphi_b} - q\sigma_0^2/(2kT), \tag{1}$$

and that between n and T is

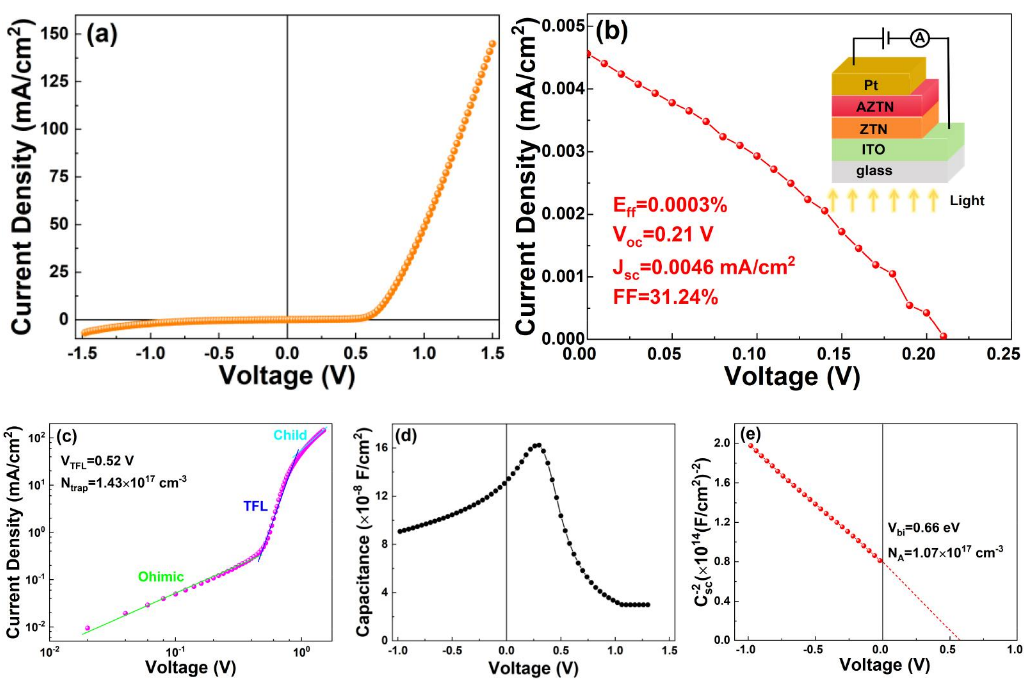

FIG. 1. The J–V curve of the homojunction measured in darkness (a) and under illumination (b). (c) Double-logarithmic J–V plot; (d) The C–V curve. (e) The C_{SC}^{-2}–V curve.

$$n^{-1} - 1 = -\rho_2 + q\rho_3/(2kT), \tag{2}$$

where k is the Boltzmann constant, $\overline{\varphi_b}$ and σ_0 are the standard barrier height and deviation at zero-bias while ρ_2 and ρ_3 are two coefficients (Fig. S7). The effective Richardson constant (A^{**}) obtained from the modified Richardson plot is 14.58 A cm^{-2} K^{-2}, near the theoretical[14] 14.40 A cm^{-2} K^{-2}.

The J–V curves of the solar cell can also be described by the single-diode equation,[39]

$$J = J_0\left\{\exp\left[q(V - JR_s)/(nkT)\right] - 1\right\} + GV - J_L, \tag{3}$$

where J_0 is the diode current density, G is the shunt conductance, and J_L is the photo-induced current density ($J_L = 0$ under darkness). The four parameters J_0, R_s, n, and G can be extracted with the analytic method[39] and those from the JV curve at 300 K are in Fig. S8. All the extracted data are in Table S1. Compared with the Cu_2ZnSnS_4/ZTN heterojunction,[19] the homojunction has higher R_s, indicating restricted carrier transport, possibly due to the relatively higher density of defect states. From the plot of ln J_0 vs 1/(kT) (Fig. S9), the current is also controlled by thermionic emission below 320 K and above that by GR in the space charge region.

Figure 2(a) shows that the Richardson plot is not linear in 210–320 K. Applying Eq. (2) to the ideality factors n in this region yields $\rho_2 = -0.06$ and $\rho_3 = -0.03$V [Fig. 2(b)]. In the single-diode model, J_0 is written as[39–44]

$$J_0 = J_{00}\exp[-q\varphi_b/(nkT)], \tag{4}$$

where J_{00} depends on the specific recombination mechanism. From Eq. (4), φ_b at each temperature can be calculated and listed as φ_{b1} in Table S1 since J_0 and n are known while J_{00} actually equals $A^{**}T^2$ in 210–320 K due to the fact that the dark forward current is dominated by TE in this range. The obtained φ_{b1} is much larger than those obtained from the Cheungs' method. In addition, applying Eq. (1) to φ_{b1} [Fig. 2(c)] gives negative ${\sigma_0}^2$, which is unphysical. From the modified Richardson plot based on the negative σ_0^2 [Fig. 2(d)], the extracted A^{**} is about 10^{-27} A cm^{-2} K^{-2}, far smaller than the theoretical value.

If the ideality factor n is omitted from Eq. (4), J_0 in the single-diode model will be

$$J_0 = J_{00}\exp[-q\varphi_b/(kT)]. \tag{5}$$

φ_b at each temperature can also be calculated and listed as φ_{b2} in Table S1. The obtained φ_{b2} is near to those obtained with the Cheungs' method. Applying Eq. (4) to φ_{b2} [Fig. 2(e)] yields $\sigma_0^2 = 0.026$ V^2 and $\overline{\varphi_{b2}} = 1.13$ V. In Fig. 2(f), the modified Richardson plot based on $\sigma_0^2 = 0.026$ V^2 not only yields $\overline{\varphi_{b2}} = 1.13$ V but also A^{**} of 13.46 A cm^{-2} K^{-2}, which is near the theoretical value. Therefore, when the dark forward current is governed by TE in the single-diode model, the barrier height calculated from the diode current J_0 excluding the ideality factor n meets the Gaussian distribution model.

The n–T relation[45,46] under nonradioactive interface recombination (IR) with tunneling enhancement is

$$n = E_{00}/(kT)\coth[E_{00}/(kT)], \tag{6}$$

where E_{00} is the characteristic tunneling energy while that under nonradiative bulk recombination (BR) with tunneling enhancement is

$$n^{-1} = \frac{1}{2}\left[1 - \frac{E_{00}^2}{3(kT)^2} + \frac{T}{T^*}\right], \tag{7}$$

where kT^* is the characteristic energy of the distribution of trap states near the band edge associated with band-tailing or potential fluctuation. Figure 3(a) shows the least squares method fitting of the ideality factors in Table S1 to Eq. (6) with $E_{00} = 58.70$ meV, and to Eq. (7) with $E_{00} = 27.40$ meV and $kT^* = 69.13$ meV. The top dashed line in Fig. 3(a) is from Eq. (7) with $E_{00} = 0$ meV and $kT^* = 69.13$ meV. Below 280 K, Eq. (6) gives a better fitting, indicating that below this

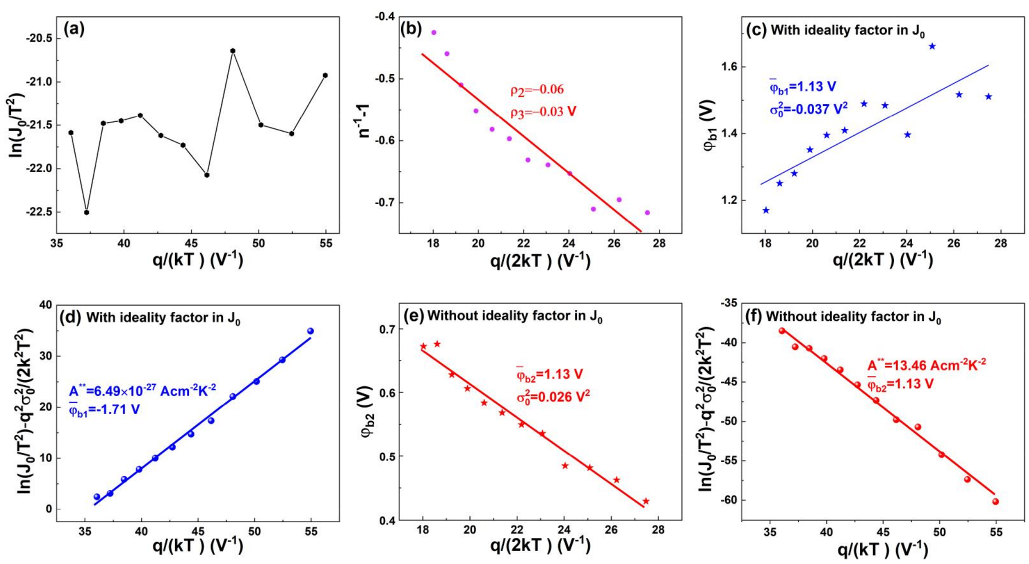

FIG. 2. (a) The curve of $\ln(J_0/T^2)$ vs q/(kT). (b) The curve of $n^{-1} - 1$ vs q/(2kT). The curve of φ_{b1} vs q/(2kT) plot (c) and the modified Richardson plot (d) (both with n in J_0). The curve of φ_{b2} vs q/(2kT) (e) and the modified Richardson plot (f) (both without n in J_0).

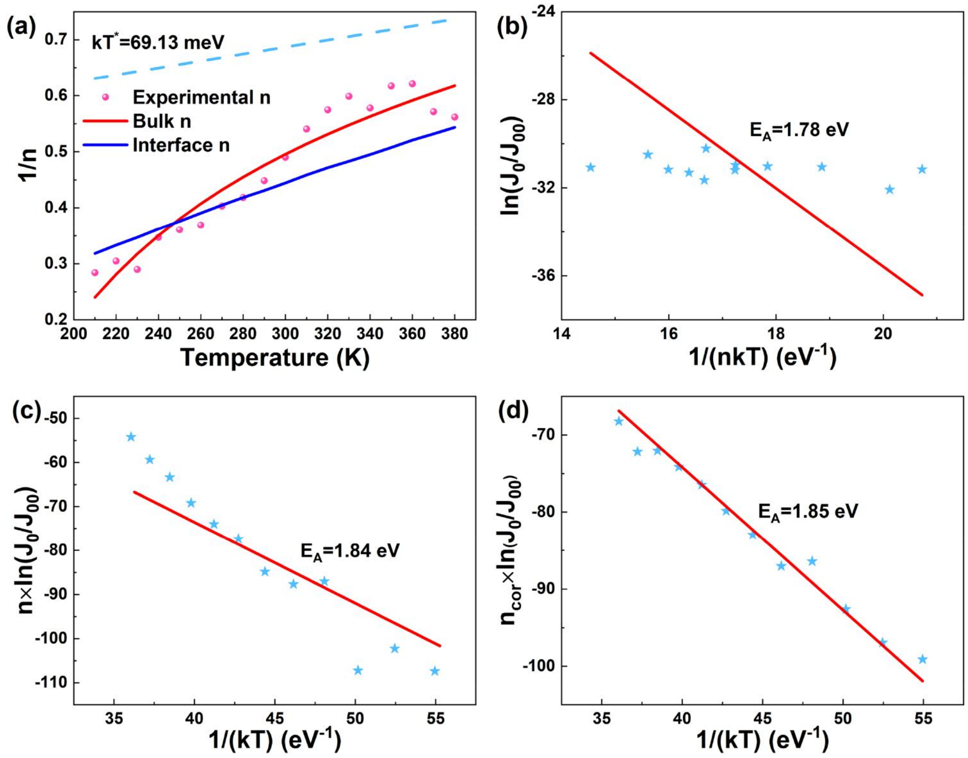

FIG. 3. (a) Fitting the ideality factor to Eqs. (6) and (7). (b) The curve of $\ln(J_0/J_{00})$ vs $1/(nkT)$. (c) The curve of $n\ln(J_0/J_{00})$ vs $1/(kT)$. (d) The curve of $n_{cor}\ln(J_0/J_{00})$ vs $1/(kT)$.

temperature, tunneling enhanced IR is dominant. The effective defect density N_{def} is calculated to be $1.32\times10^{19}\ \mathrm{cm}^{-3}$, from $E_{00} = hN_{def}^{0.5}/[2(m^*\varepsilon_r\varepsilon_0)^{0.5}]$, where h is the Plank constant and m^* is the effective electron mass (m^* is 0.12 times that of a rest electron[47–50]) This high N_{def}, which is from all the interface states that can provide the path for tunneling, is in agreement with the poor crystallization of ZTN and AZTN, leading to the relatively low J_{sc}.[46] The fitting to Eq. (7) is roughly acceptable in the whole range but better than Eq. (6) in the higher temperature range, indicating that tunneling enhanced BR gradually gets dominant at higher temperatures.

To obtain the activation energy E_A of the electrical transport below 320 K, Eq. (4) is rewritten as[46]

$$J_0 = J_{00}\exp[-E_A/(nkT)]. \tag{8}$$

Figure 3(b) shows the curve of $\ln(J_0/J_{00})$ vs $1/(nkT)$, hardly any activation energy can be obtained while Fig. 3(c) shows the curve of $n\ln(J_0/J_{00})$ vs $1/(kT)$ with poor linearity observed. To solve these problems, the corrected ideality factor n_{cor} is calculated according to Eq. (6) with $E_{00} = 58.70$ meV. Figure 3(d) shows the curve of $n_{cor}\ln(J_0/J_{00})$ vs $1/(kT)$, good linearity is observed. The obtained E_A is 1.85 eV, smaller than the bandgap of *n*-type ZTN, also implying IR dominance in this region.

Under IR, V_{oc} is controlled by the interfacial barrier height φ_H of the holes, and the theoretical V_{oc} at 300 K is 1.49 V (Fig. S5), much larger than the observed 0.21 V. V_{oc} controlled by IR and affected by barrier height inhomogeneity is[39,42,44] $V_{oc} = n\overline{\varphi_b} - qn\sigma_0{}^2/(2kT) - (nkT/q)\ln(qS_{it}N_V/J_L)$ according to Eqs. (3) and (5), where S_{it} is the IR velocity. With the measured V_{oc}, J_L and the parameters obtained from the single-diode model, the calculated S_{it} is 1.77×10^4 cm/s, near that of other semiconductors.[42] In addition, V_{oc} according to Eqs. (3) and (8) is[39,42,44] $V_{oc} = [\varphi_{b1} - q\sigma^2/(2kT)] - (nkT/q)\ln(qS_{it}N_V/J_L)$, where $\sigma = 0.03$ V is the standard deviation of φ_{b1} (Fig. S10). From this, S_{it} at 300 K is calculated to be 1.36×10^4 cm/s, with $q\varphi_{b1} = 1.28$ eV at 300 K. Therefore, strong IR as well as barrier height inhomogeneity or the temperature fluctuation of interface barrier heights for holes results in low V_{oc}.

The current in 330–380 K is well fitted to the current formula due to GR in the space charge region with deep energy levels in the middle of the forbidden bandgap[51,52] (Fig. S11). This suggests the GR in the space charge region is mainly due to deep energy levels in the middle of the forbidden bandgap.

The double-logarithmic J–V curves in 300–380 K under 1.0–1.5 V are in Fig. 4(a). These curves exhibit typical SCLC behavior, indicating that charge transport in the AZTN layer is dominated by a single trap level. In this regime, the current density is $J = 9\varepsilon_r\varepsilon_0\mu_p\theta V^2/(8d^3)$,[53] where θ is the trap factor. The values of θ at various temperatures can be extracted from the intercepts on the vertical axis in Fig. 4(a). The relation between the trap factor θ and the defect energy level E_t measured from the valence band maximum (E_V) is $\theta = (N_V/N_t)\exp[-E_t/(kT)]$, where N_V is the effective density of states in the valence band, and N_t is the concentration of traps at this energy level. By plotting $\ln\theta$ as a function of $1/(kT)$ [Fig. 4(b)], E_t is found to be 0.13 eV. Since the carrier density of the *n*-type ZTN is larger than that of *p*-type AZTN, E_t can be considered to be in the p side and in close proximity to E_v. This indicates it is a shallow acceptor level, most likely resulting from substituting Ag for Zn or Sn [Fig. 4(e)].

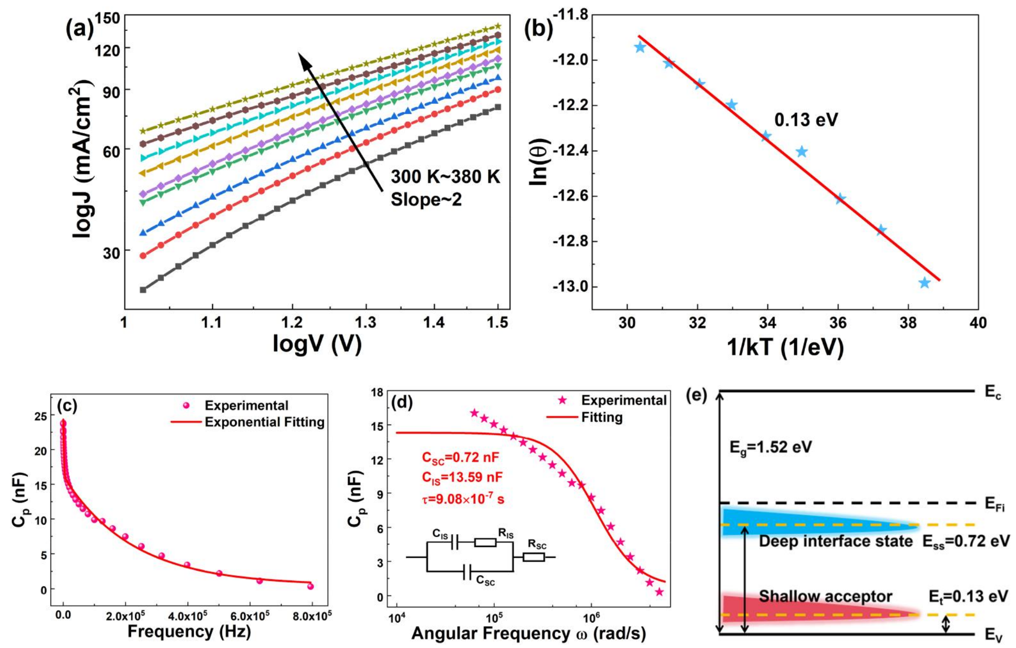

FIG. 4. (a) Double-logarithmic J–V curves in the SCLC regime in 300–380 K. (b) The curve of ln θ vs 1/(kT). (c) The C_p–f curves and the exponential fitting. (d) The fitting to the C_p–f curve (inset: the equivalent circuit for fitting). (e) The energy levels.

Figure 4(c) shows the zero-bias admittance spectroscopy at 300 K. The capacitance (C_p) exhibits a strong frequency (f) dependence over the measured range. Fitting the C_p-f data with an exponential decay function [solid line in Fig. 4(c)] confirms that C_p decays exponentially with increasing frequency. This behavior is typically attributed to the dispersion effect of interface states: under high-frequency conditions, deep-level traps in quasi-equilibrium with the semiconductor Fermi level cannot respond sufficiently to the alternative current (AC) signal, resulting in a decrease in the measured capacitance.[54–56] This frequency dependence can be described by an equivalent circuit model incorporating interface state response, in which[54] $C_p = C_{SC} + C_{IS}/[1 + (\omega\tau)^2]$, where $\omega = 2\pi f$ is the angular frequency and τ is the interface trap time constant of the interface states. The C_p-f data are fitted to this equation using the least squares method with C_{SC}, C_{IS}, and τ as the fitting parameters [Fig. 4(d)] and the fitting yields $C_{SC} = 0.72$ nF, $C_{IS} = 13.59$ nF, and $\tau = 9.08 \times 10^{-7}$ s. The interface state density $N_{ss} = C_{IS}/(qA_j)$ is calculated to be $5.31 \times 10^{11}\,\mathrm{eV^{-1}cm^{-2}}$ ($A_j = 0.16\,\mathrm{cm^2}$). The interface state energy level E_{SS} satisfies $E_{SS} - E_V = q(\varphi - V)$, where $\varphi = V_{bi} + kT/q\ln(N_V/N_A)$. Since the C_p–f measurement is performed at V = 0 V while $V_{bi} = 0.66$ V, substituting $N_V = 1.04 \times 10^{18}\,\mathrm{cm^{-3}}$ and $N_A = 1.07 \times 10^{17}\,\mathrm{cm^{-3}}$ yields $\varphi = 0.72$ eV. Thus, $E_{SS} \approx E_V + 0.72$ eV, indicating that the interface state is located approximately 0.72 eV above E_V of the AZTN layer, corresponding to a deep-level defect [Fig. 4(e)].

The origin of this deep level is speculated based on the results in the literature. First, shallower isolated defects such as Sn_{Zn} (about 0.37 eV below the conduction band minimum E_c) and O_N (E_c–0.21 eV) can be ruled out.[6] Theoretical studies suggest that under Zn-rich and unintentional oxygen-containing conditions, the charge-neutral complex (Zn_{Sn} + $2O_N$) is energetically dominant.[6,57] Nevertheless, in the cation-disordered interface environment formed during actual deposition, Zn_{Sn} and O_N may not pair ideally, potentially leading to acceptor-type complexes such as (Zn_{Sn} + O_N). Owing to incomplete charge compensation, such complexes may introduce deep levels near the midgap. In addition, the doped Ag atoms may also result in deep energy levels by forming different complexes with neighboring intrinsic vacancies or antisite defects.

Figure 5(a) shows the depth profile of the charge density N_{CV} calculated from the C–V measurements and N_{DLCP} calculated from drive-level capacitance profiling (DLCP) measurements at 10 kHz. The variation of N_{CV} with depth X_{CV} ($X_{CV} = \varepsilon_r\varepsilon_0 A_j/C$) is given by $N_{CV} = C^3(dC/dV)^{-1}/(q\varepsilon_0\varepsilon_r A_j^2)$.[19] The variation of N_{DLCP} with depth X_{DLCP} is given by[58] $N_{DLCP} = -C_0^3/(2q\varepsilon_r\varepsilon_0 A_j^2 C_1)$, where C_0 and C_1 are the fitting parameters during the DLCP measurement while $X_{DLCP} = \varepsilon_r\varepsilon_0 A_j/C_0$. Since N_{CV} includes contributions from free carriers, bulk defects, and interface states, whereas N_{DLCP} results only from the former two, the interface state density N_i is $7.86 \times 10^{16}\,\mathrm{cm^{-3}}$ (N_{CV}-N_{DLCP}). The extracted N_i here is smaller than N_{def} from Fig. 3(a) since N_i here is just from those that can respond to the measurement frequency. It is higher as compared with other ZTN-based junctions,[19,20] resulting in lower E_{ff}.

From the Nyquist plot of the homojunction [Fig. 5(b)], a single semicircular arc is observed. The inset illustrates the equivalent circuit used for fitting, which includes the series resistance R_s, the recombination resistance R_{rec} and the parallel capacitance C. The fitting yields $R_s = 44.77\ \Omega$, $R_{rec} = 132.50\ \Omega$, C = 14.58 nF, and shows good agreement with the experimental data. The recombination time constant (CR_{rec}) is about 1.93 μs, shorter than other ZTN-related solar

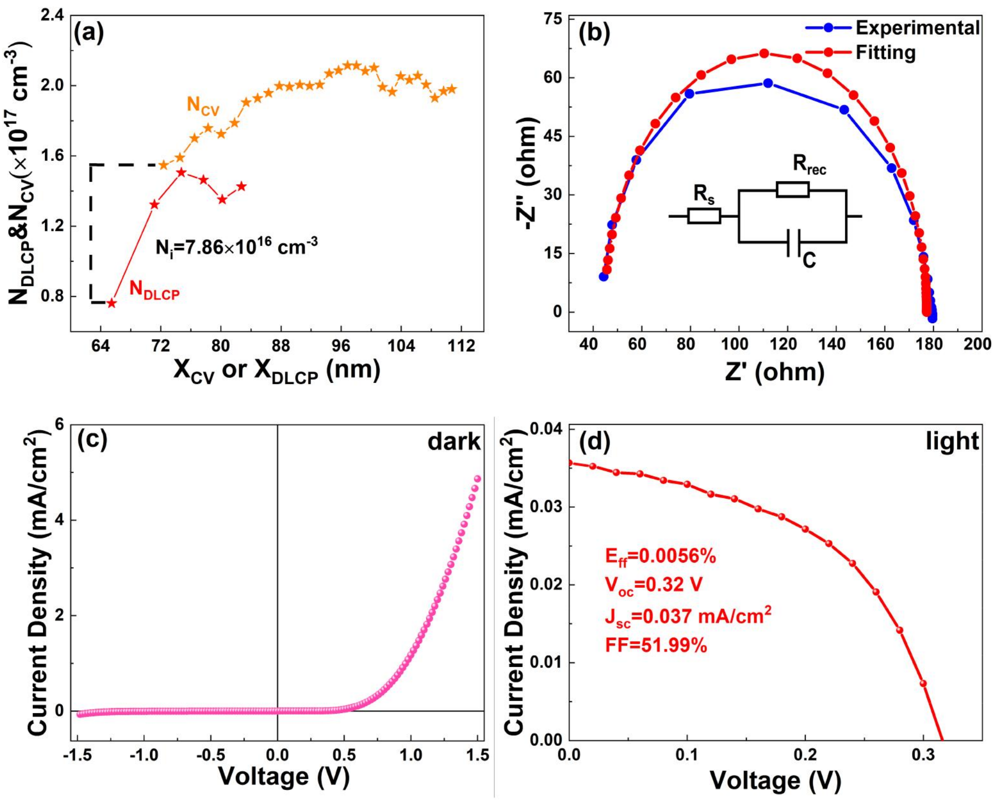

FIG. 5. (a) Extracting the interface state density. (b) The Nyquist plot (inset: the equivalent circuit for fitting). The dark (a) and illuminated (b) J–V curve of the homojunction with thinner ZTN and thicker AZTN.

cells,[19,20] and implying higher minority carrier recombination rate, mainly due to the lower R_{rec}, which is another factor[59] leading to the relatively low E_{ff} of the present homojunction.

By reducing the deposition time of ZTN and increasing that of AZTN (detailed in supplementary material), the photovoltaic performance can be improved [Figs. 5(c) and 5(d)], demonstrating the promise for improving the performance.

In conclusion, the properties of AZTN/ZTN homojunction solar cells prepared by magnetron sputtering are studied, and the mechanism that limits the photovoltaic performance is revealed. The abrupt homojunctions exhibit pronounced rectification under dark conditions and a relatively weak photovoltaic response under illumination. The current transport is dominated by TE in 190–320 K and GR in the space charge region at higher temperature. Its J_{sc} is limited mainly by the high effective defect density and IR. In addition to IR, its V_{oc} is also affected by barrier height inhomogeneity or the temperature fluctuation of interface barrier height for holes. A shallow acceptor level at $E_V + 0.13$ eV and a deep interface state level at $E_V + 0.72$ eV are observed. The relatively high interface state density and low recombination resistance also limit the E_{ff}. The promise for improving the photovoltaic performance of the homojunctions is demonstrated.

See the supplementary material for details on the following: preparing and characterizing the films, including ZTN, AZTN, and Pt and the homojunctions; the steps for preparing the homojunctions, XRD patterns, and SEM images; the reflectance, transmittance, and Tauc plots; the energy band diagram; analyzing the current transport mechanism; analyzing the barrier height inhomogeneity with the Gaussian distribution model; extracting the parameters in the single-diode model; the parameters extracted with the single-diode model and the calculated barrier heights; the curve of $\ln J_0$ vs $1/(kT)$; obtaining the standard deviation σ for φ_{b1}; fitting the forward-bias currents in 330–380 K; the detail about the new homojunctions in which the deposition time of ZTN is 40 min and that of AZTN is 4 h.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Fan Ye: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Validation (equal);

Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Qian Gao:** Data curation (equal); Formal analysis (equal); Investigation (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Qiao Tang:** Data curation (supporting); Investigation (supporting). **Yi-Zhu Xie:** Resources (equal). **Ying Liu:** Resources (equal). **Chun-Yan Yu:** Resources (equal). **Fan Wang:** Resources (equal). **Dong-Ping Zhang:** Resources (equal). **Xing-Min Cai:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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